

Kimberly Wang

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EDUCATION

Rensselaer Polytechnic Institute

Troy, NY

B.S. Electrical Engineering, GPA: 3.71

Aug 2024 – Dec 2027

Completed Coursework: Electric Circuits, Computer Components and Operations, Embedded Control, Intro to Logic Based AI, Intro to Electronics (Spring 2026), Intro to Engineering Design (2026)

TECHNICAL SKILLS

Hardware: Arduino, MSP432, Basys 3, Soldering, Oscilloscope, Waveform Generator, Multimeter, Breadboard

Software: LTspice, Cadence, Altium Designer, Vivado, Siemens NX, LabVIEW, MATLAB, Simulink, Office Suite

Languages: Python, C, VHDL

EXPERIENCE

Undergraduate Researcher @ RPI Mercer XLab

Jan 2025 – Present

Rensselaer Polytechnic Institute

Troy, NY

Software Configured Breadboard (SCB) Supervised by ADI Fellow Douglas Mercer

- Used Python and ALICE software to computerize the wiring of breadboards to reduce debugging and wiring issues
- Executed board-level validation and root-cause analysis utilizing automated Python scripts and a multimeter
- Contributed to hardware design reviews, providing feedback on the PCB layout and component choices
- Authored technical documentation and user guides to streamline the onboarding process for new users
- Collaborating with RPI's AI club to utilize Cursor AI to automate circuit design, building, and testing

Undergraduate Assistant for "Intro to ECSE" Course

Aug. 2025 – Present

Rensselaer Polytechnic Institute

Troy, NY

- Help students in debugging and implementing C programming on Arduino and MATLAB
- Advise on analog and digital circuit building, testing, and measurement
- Clarify questions about theoretical concepts within contexts of labs

PROJECTS

CO Sensing and Ventilating System | *LTspice, Oscilloscope, Waveform Generator*

Sep 2025 – Present

- Created a multi-stage analog system for maintaining safety from CO in buildings
- Designed and built the initial indication stage using an RC filter and comparator with an LED output
- Configured autonomous fan controls to trigger active ventilation controls upon the indication signal
- Utilized first and second order filters and 555 Timers to achieve desired signal inputs
- Prototype final stage of an audible alerting sub-system for added safety

LED Sequencer | *Altium Designer*

Nov 2025

- Designed a custom PCB in Altium Designer following manufacturing constraints on trace widths and clearance
- Generated manufacturing deliverables (Gerbers, BOM) and selected components based on vendor specifications

Compass Controlled TI RSLK Car | *MSP432, C, I2C*

Apr 2025

- Designed a closed-loop navigation system using a magnetometer via I2C protocols for real time directions
- Implemented PID controls in C to minimize errors in motor movements
- Configured GPIO interrupts to handle data acquisition and maintain stable motor control

PI Controller Design | *LTspice, Oscilloscope*

Nov 2024

- Designed and simulated an analog PI controller in LTspice to achieve a steady state voltage in 4s
- Verified the design by building and testing the circuit to get results with < 5% error from simulation